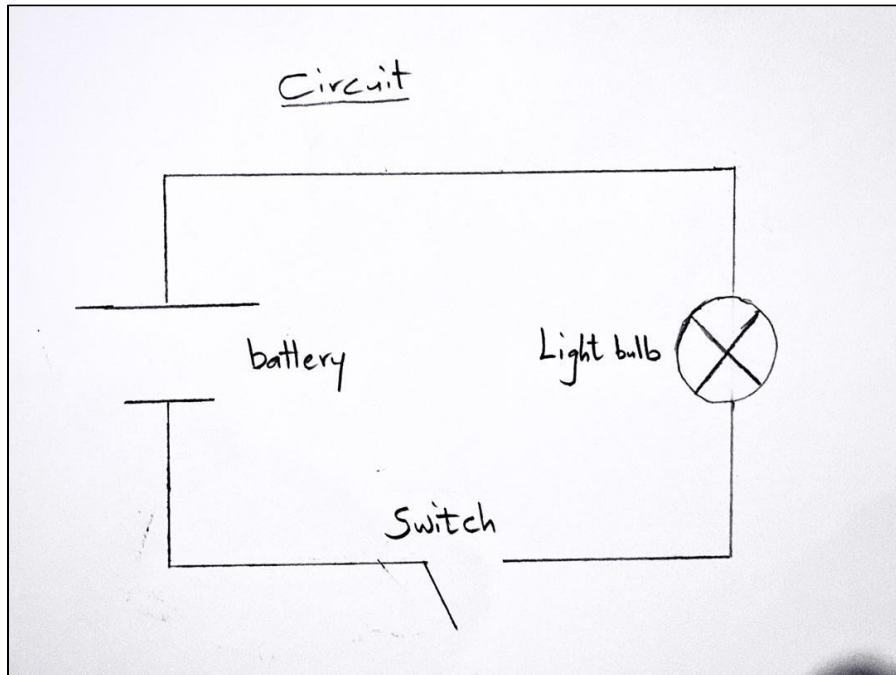


Computer Architecture
Assignment 2 (49m)
(ANSWER ALL QUESTIONS)
CLO2-PLO2-C2

1. Refer to the video **Boolean Logic & Logic Gates** (<https://www.youtube.com/watch?v=gI-qXk7XojA>) and answer the following questions:
 - a) What 0 and 1 represents?
 - b) Draw the simplest circuit.
 - c) Make a Boolean table for OR, AND, NOT and XOR.
 - d) Draw and explain how OR logic gate works?
 - e) Draw and explain how XOR logic gate works?
 - f) Draw all the symbols of all the four logic gates.

1. a) 0 represents off or false while 1 represents on or true.

b)



c) OR

a	b	out
0	0	0
0	1	1
1	0	1
1	1	1

AND

a	b	out
0	0	0
0	1	1
1	0	1
1	1	1

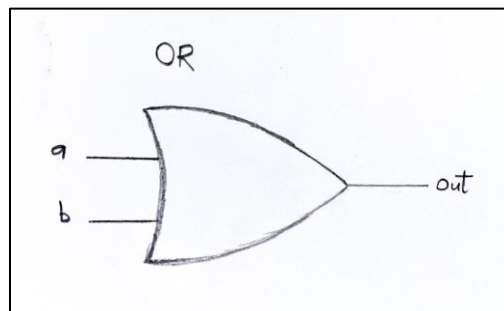
NOT

in	out
0	1
1	0

XOR

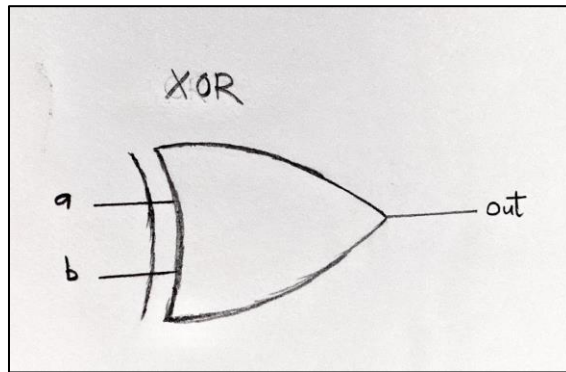
a	b	out
0	0	0
0	1	1
1	0	1
1	1	0

d)



Explanation: An OR gate gives an output of 1 if at least one input is 1. It only gives 0 when both inputs are 0.

e)

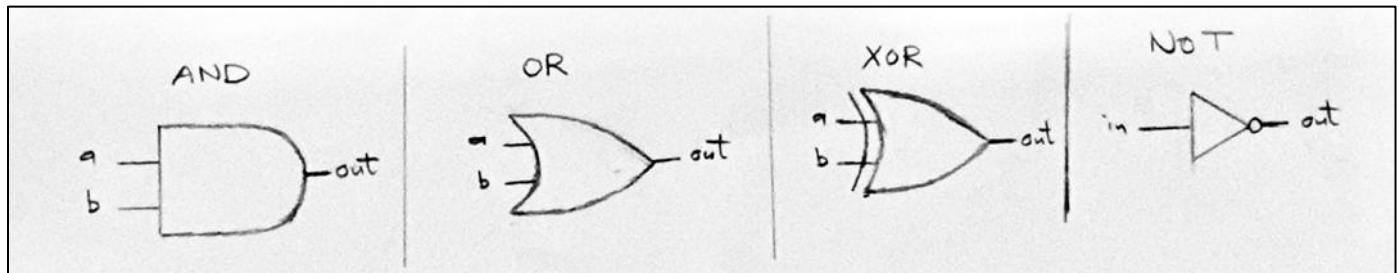


Explanation:

An XOR gate outputs 1 only when the inputs are different.

- If both inputs are the same (0,0 or 1,1), the output is 0.
- If inputs differ (0,1 or 1,0), the output is 1

f)



2. Refer to the video **Binary Numbers** (<https://www.youtube.com/watch?v=1GSjbWt0c9M>) and answer the following questions:

- The regular number we always use in everyday life is base-_____ number?
- What is base-two notation?
- Write 1-8 in binary.
- What is the result of the following binary problems?
Show your calculation for: $0+0$; $1+0$; $1+1$; $10+1$; $11+1$
- Instruction format of 32-bit floating point number? Explain each part.
- How computer represents text or alphabets?
- What is the main difference between ASCII and Unicode?

Answer

- a) The regular number we always use in everyday life is base- 10 number.
b) to represent data because transistors can represent two states: 1 and 0.

c)

Decimal	Binary
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000

- d) $0 + 0 = 0$
 $1 + 0 = 1$
 $1 + 1 = 10$
 $10 + 1 = 11$
 $11 + 1 = 100$

e)

Part	Bits	Explanation
Sign	1	0 = positive, 1 = negative
Exponent	8	Stores exponent with bias 127
Mantissa (Fraction)	23	Fractional part (with implicit leading 1)

- f) Computers represent text and alphabets using binary codes, with the most common system being ASCII (American Standard Code for Information Interchange).

g)

Feature	ASCII	Unicode
Characters	128(Basic English letters & symbols)	Over 143,000 (Global language support)
Bit size	7 bits (commonly stored in 8 bits)	Typically uses 8, 16, or 32 bits
Language support	Only English	Multilingual: Chinese, Arabic, emojis & more
Era	Created in the 1960s	Created in the 1990s
Encoding formats	Single format	Multiple: UTF-8, UTF-16, UTF-32

3. Refer to the video **Pipelining** (<https://www.youtube.com/watch?v=ecCt6HPIPeA>) and answer the following questions:
- a) What is the main difference between unpipelining and pipelining in CPU.
 - b) What is the advantage of pipelining?
- a) The main difference between unpipelining and pipelining in CPU is:
- An unpipelined CPU executes one instruction at a time, completing all stages (fetch, decode, execute, etc.) before starting the next.
 - Pipelined CPU allows multiple instructions to be in different stages at the same time, increasing overall speed and efficiency.
- b) Pipelining allows multiple instructions to be processed at the same time, each at a different stage. This improves CPU performance by executing more instructions in less time, making the processor faster and more efficient.